Group 17 Final Presentation: Organ donor registry

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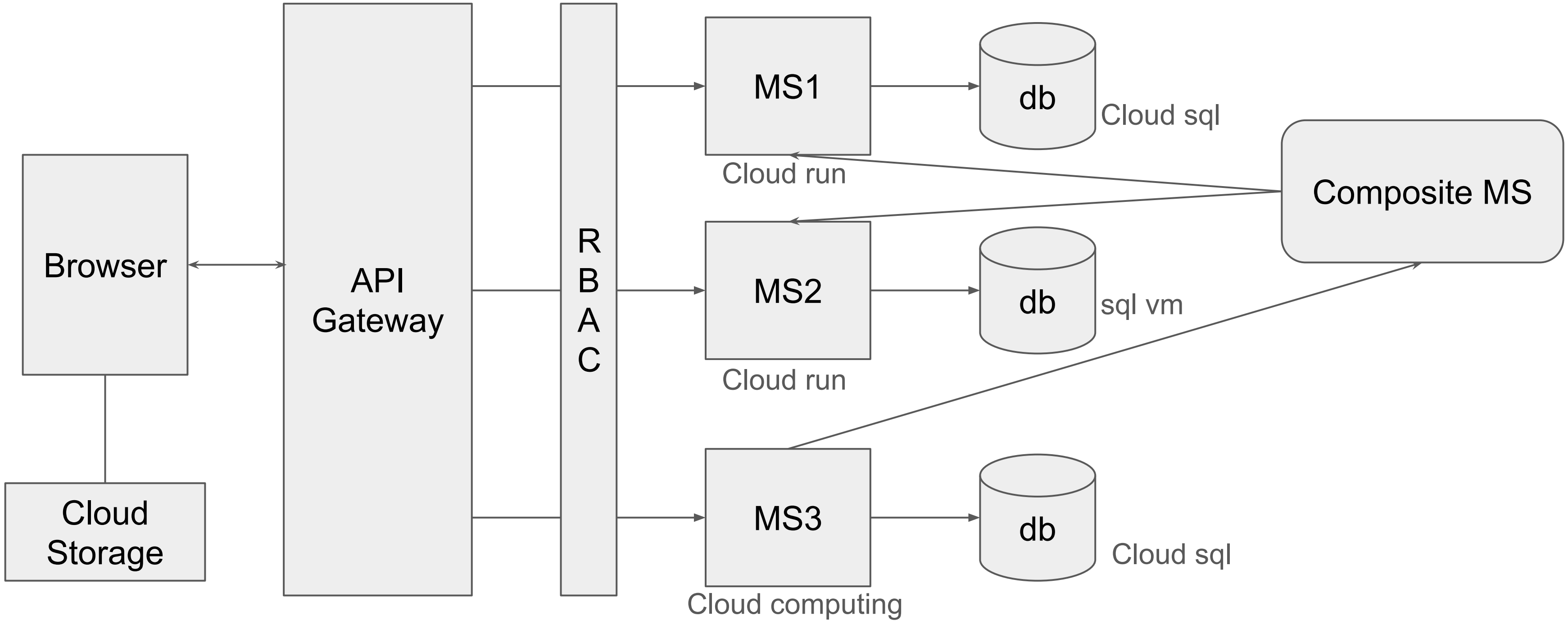
Project Overview

Content:

We built an organ donation management system using a microservices architecture.

- Manages donor information and available organs.
- Tracks hospitals, recipients, and their organ needs.
- Performs matching between donors and recipients.

Project Architecture



Microservice 1: Donor Registry

Typical Flow

1. A **Donor** is registered in the system.
2. The Donor’s available **Organs** are recorded.
3. The Donor’s **Consent** for donation is captured.
4. The Organ Matching Service (MS3) uses this data to perform organ matching with recipient needs from MS2.

MS1: Open API

Cloud Run URL:
https://donor-registry-service-730071231868.us-central1.run.app/docs

Github Repository:
https://github.com/TheSkyRS/Donor-Registry-Service

Docker hub:
https://hub.docker.com/repository/docker/yonghaolin/donor-registry-service

donors

- GET `/donors` List Donors
- POST `/donors` Create Donor
- GET `/donors/{donor_id}` Get Donor
- PUT `/donors/{donor_id}` Update Donor
- DELETE `/donors/{donor_id}` Delete Donor
- GET `/donors/{donor_id}/organs` List Organs For Donor
- POST `/donors/{donor_id}/organs` Create Organ For Donor
- GET `/donors/{donor_id}/consents` List Consents For Donor
- POST `/donors/{donor_id}/consents` Create Consent For Donor

organs

- GET `/organs` List Organs
- GET `/organs/{organ_id}` Get Organ
- PUT `/organs/{organ_id}` Update Organ
- DELETE `/organs/{organ_id}` Delete Organ

consents

- GET `/consents` List Consents
- GET `/consents/{consent_id}` Get Consent
- PUT `/consents/{consent_id}` Update Consent
- DELETE `/consents/{consent_id}` Delete Consent

MS1: API Endpoint Details

✅ SQLAlchemy persistence (MySQL)

✅ Actual CRUD for donors, organs, consents

✅ Async FastAPI endpoints

✅ Query parameters for all collection resources.

✅ Linked data / HATEOAS-style _links

✅ Pagination for all top-level resources

✅ Correct 201 Created + Location headers

✅ ETag + If-Match version control for donors

🌐 API Surface (Sprint 2 — fully implemented)

Donors

Method	Path	Description
GET	`/donors`	List donors (filter by `name`, `blood_type`, `status`; pagination)
POST	`/donors`	Create donor (`201 Created` + `Location` header)
GET	`/donors/{id}`	Retrieve donor by ID (includes `ETag`)
PUT	`/donors/{id}`	Update donor (**requires** `If-Match` ETag)
DELETE	`/donors/{id}`	Delete donor

Organs

Method	Path	Description
GET	`/organs`	Filter by `donor_id` / `organ_type`, pagination
GET	`/organs/{id}`	Retrieve organ by ID
PUT	`/organs/{id}`	Update organ
DELETE	`/organs/{id}`	Delete organ
GET	`/donors/{donor_id}/organs`	List organs belonging to a donor
POST	`/donors/{donor_id}/organs`	Create organ for donor

Consents

Method	Path	Description
GET	`/consents`	Filter by `donor_id` / `status`, pagination
GET	`/consents/{id}`	Retrieve consent by ID
PUT	`/consents/{id}`	Update consent
DELETE	`/consents/{id}`	Delete consent
GET	`/donors/{donor_id}/consents`	List consents for donor
POST	`/donors/{donor_id}/consents`	Create consent for donor

MS1: API endpoints Features

🔗 Linked Data (HATEOAS)

Each donor response includes:

```
"_links": {
  "self": "/donors/123",
  "organs": "/organs?donor_id=123",
  "consents": "/consents?donor_id=123"
}
```

📄 Pagination

Supported on `/donors`, `/organs`, `/consents`:

```
?limit=50&offset=0
```

Example response:

```
{
  "items": [ /* ... */ ],
  "count": 50,
  "total": 243,
  "_links": {
    "self": "/donors?limit=50&offset=0",
    "next": "/donors?limit=50&offset=50",
    "prev": null
  }
}
```

🔍 Query parameters - Filtering Examples

```
/donors?name=bob&status=active
/organs?donor_id=<uuid>&organ_type=kidney
/consents?donor_id=<uuid>&status=granted
```

🏷️ ETag Version Control

Optimistic concurrency for donors:

ETags are derived from the entire donor payload, so any field change yields a new value.

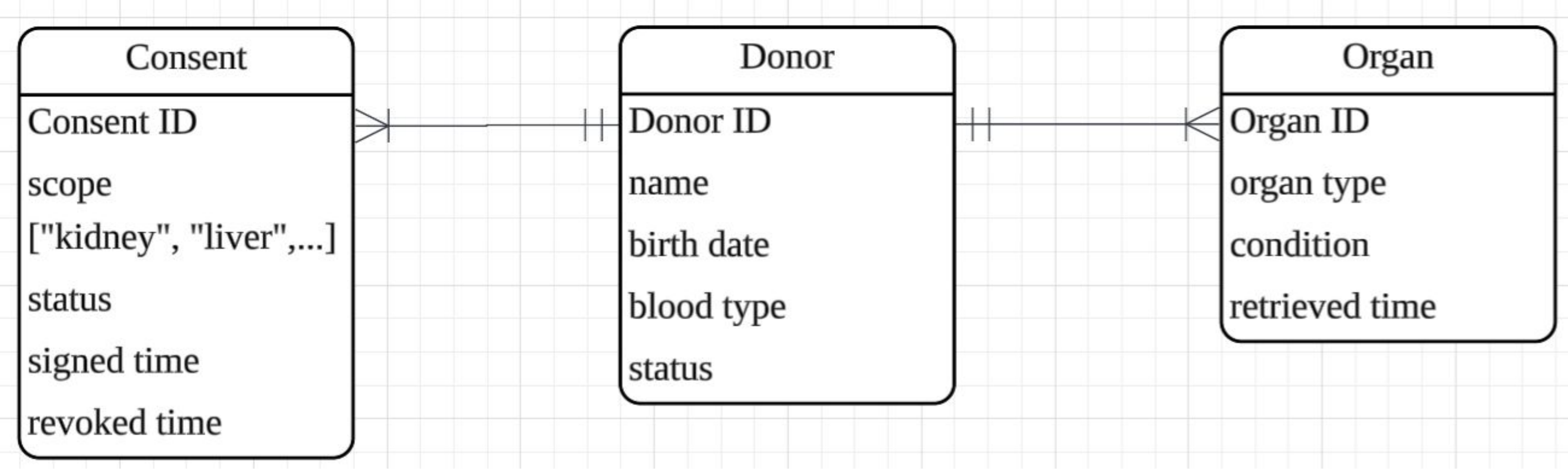
```
GET /donors/{id}
ETag: "<sha256>"

PUT /donors/{id}
If-Match: "<sha256>"
```

If mismatched:

```
412 Precondition Failed
```

MS1: Diagram



Consent
Consent ID
scope
["kidney", "liver",...]
status
signed time
revoked time
Donor
Donor ID
name
birth date
blood type
status
Organ
Organ ID
organ type
condition
retrieved time

MS1: DB model

donors

Column	Type
id	UUID
first_name	varchar(100)
last_name	varchar(100)
dob	date
blood_type	enum
status	enum
created_at	datetime
updated_at	datetime

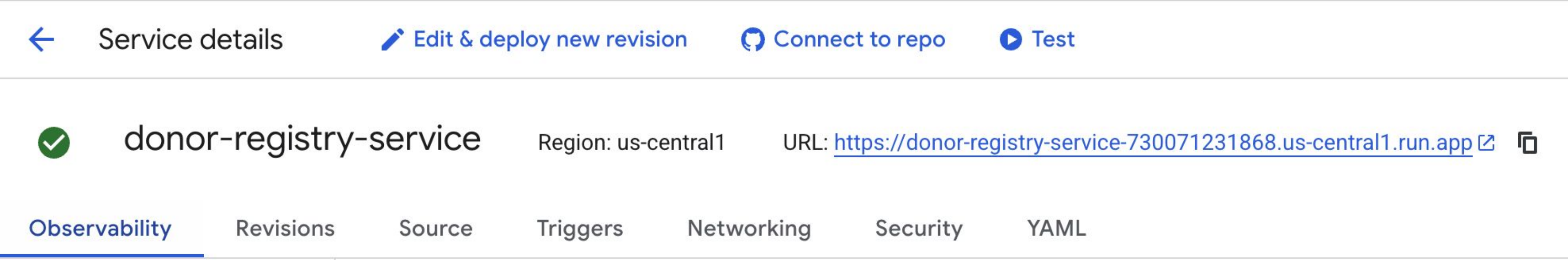
organs

Column	Type
id	UUID
donor_id	FK(donors.id)
organ_type	enum
condition	varchar(100)
retrieved_at	datetime
created_at	datetime
updated_at	datetime

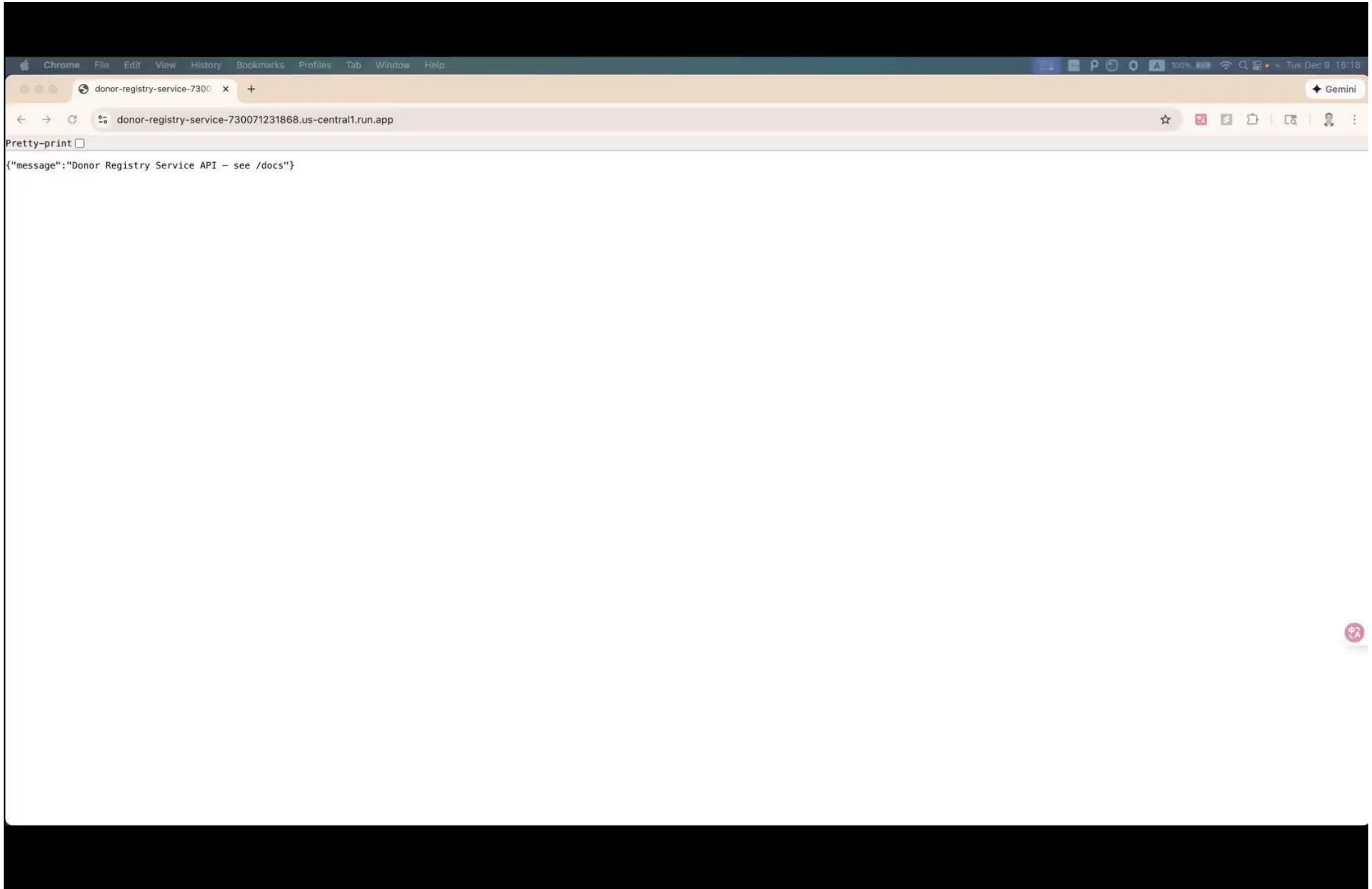
consents

Column	Type
id	UUID
donor_id	FK(donors.id)
scope	JSON
status	enum
signed_at	datetime
revoked_at	datetime
created_at	datetime
updated_at	datetime

MS1: Cloud Run and Cloud SQL consoles



MS1: Video Demo 3:53 (Required using columbia email)



Microservice 2: Recipient Waitlist

Content

Manage all data related to recipients and hospitals and to record the organ needs of patients waiting for transplantation.

Typical Flow

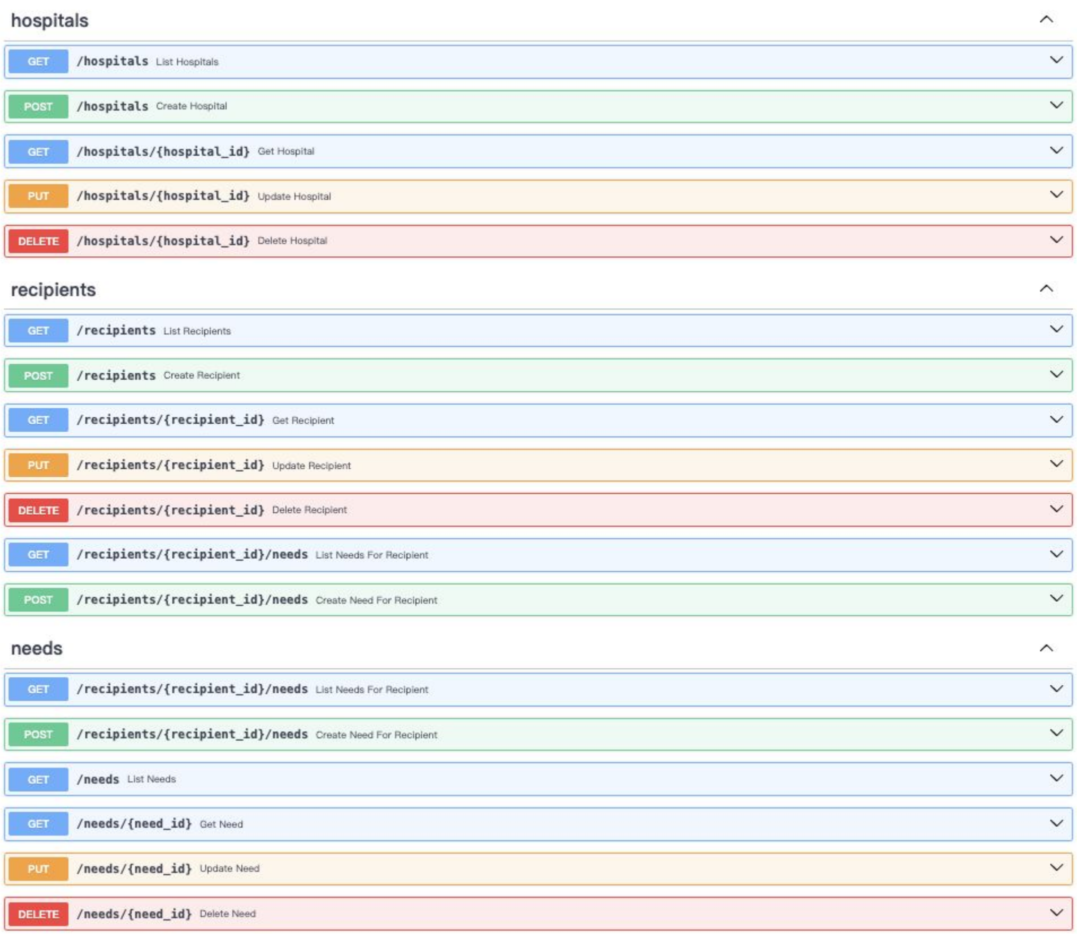
1. Register a **hospital** in the system.
2. Register a **recipient** with basic data.
3. Record **organ needs** information.
4. The Organ Matching Service (MS3) uses this data to perform organ matching with donor donation from MS1.

MS2: Open API

Cloud Run URL:
https://recipient-waitlist-service-695815011556.us-east1.run.app/docs

Github Repository:
https://github.com/zl3508/recipient-waitlist-service

Docker hub:
https://hub.docker.com/repository/docker/zhenqili/recipient-waitlist-service/general



MS2: API Endpoint Details

- ✅ MySQL persistence using mysql.connector.
- ✅ Actual CRUD for hospitals, recipients and needs.
- ✅ Nested routes for recipient needs.
- ✅ Query parameters (limit) on all collections.
- ✅ Basic domain validation (DB)

Hospitals

Method	Path	Description
GET	`/hospitals`	List (filters + `limit`)
POST	`/hospitals`	Create hospital
GET	`/hospitals/{id}`	Get hospital by ID
PUT	`/hospitals/{id}`	Update hospital
DELETE	`/hospitals/{id}`	Delete hospital

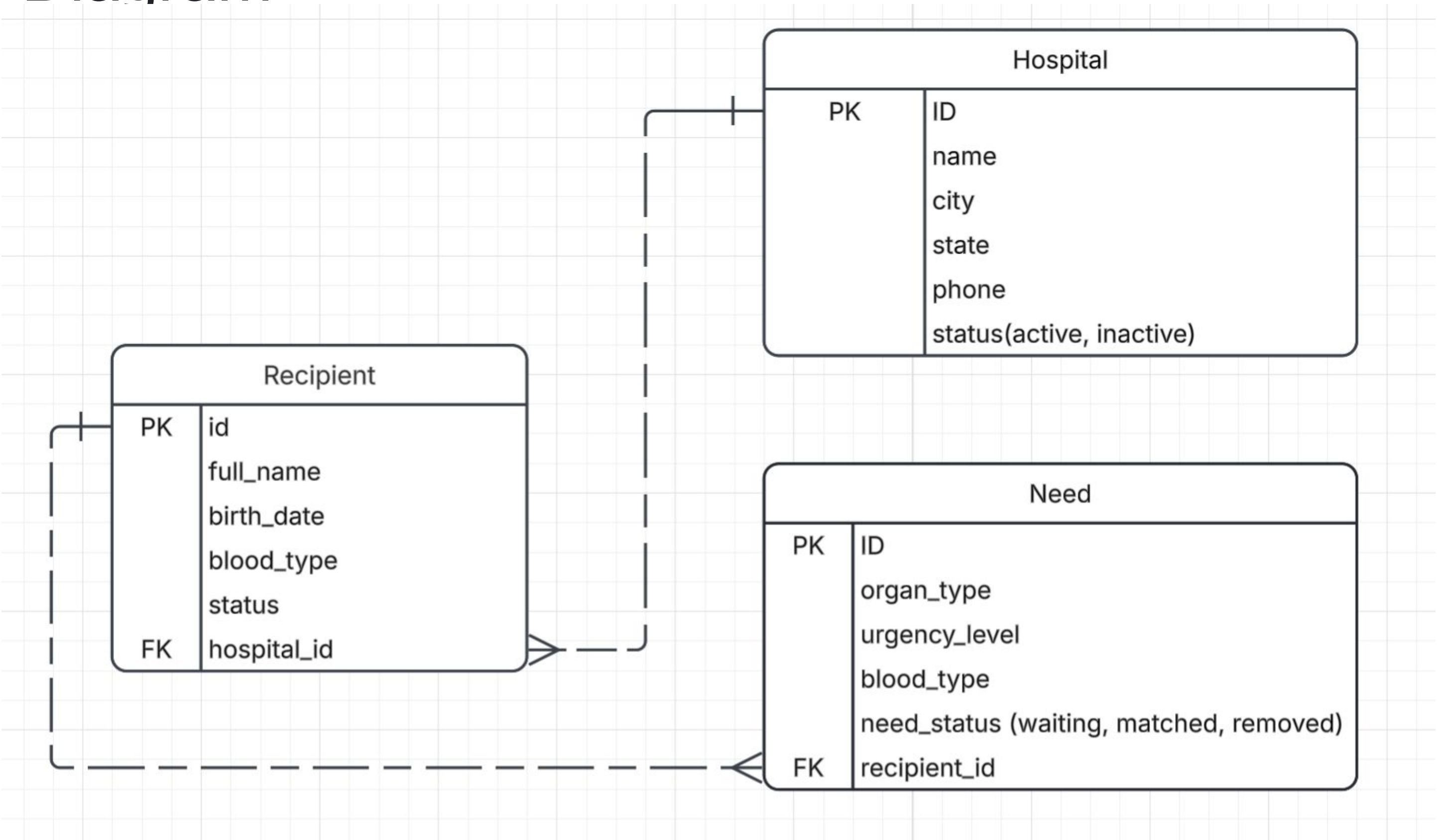
Recipients

Method	Path	Description
GET	`/recipients`	List (filters + `limit`)
POST	`/recipients`	Create recipient
GET	`/recipients/{id}`	Get recipient by ID
PUT	`/recipients/{id}`	Update recipient
DELETE	`/recipients/{id}`	Delete (blocked if needs exist)
GET	`/recipients/{recipient_id}/needs`	Needs for recipient
POST	`/recipients/{recipient_id}/needs`	Add need for recipient

Needs

Method	Path	Description
GET	`/needs`	List needs (filters + `limit`)
GET	`/needs/{id}`	Get need by ID
PUT	`/needs/{id}`	Update need
DELETE	`/needs/{id}`	Delete need
GET	`/recipients/{recipient_id}/needs`	Needs for recipient
POST	`/recipients/{recipient_id}/needs`	Add need for recipient

MS2: Diagram


Hospital
PK ID
name
city
state
phone
status(active, inactive)
Recipient
PK id
full_name
birth_date
blood_type
status
FK hospital_id
Need
PK ID
organ_type
urgency_level
blood_type
need_status (waiting, matched, removed)
FK recipient_id

MS2: DB model

Hospitals

Column	Type
id	UUID
name	varchar(128)
city	varchar(64)
state	varchar(64)
phone	varchar(32)
status	enum('active','inactive')
created_at	datetime
updated_at	datetime

Recipients

Column	Type
id	UUID
full_name	varchar(128)
dob	date
blood_type	enum
status	enum('active','inactive')
primary_hospital_id	FK(hospitals.id)
created_at	datetime
updated_at	datetime

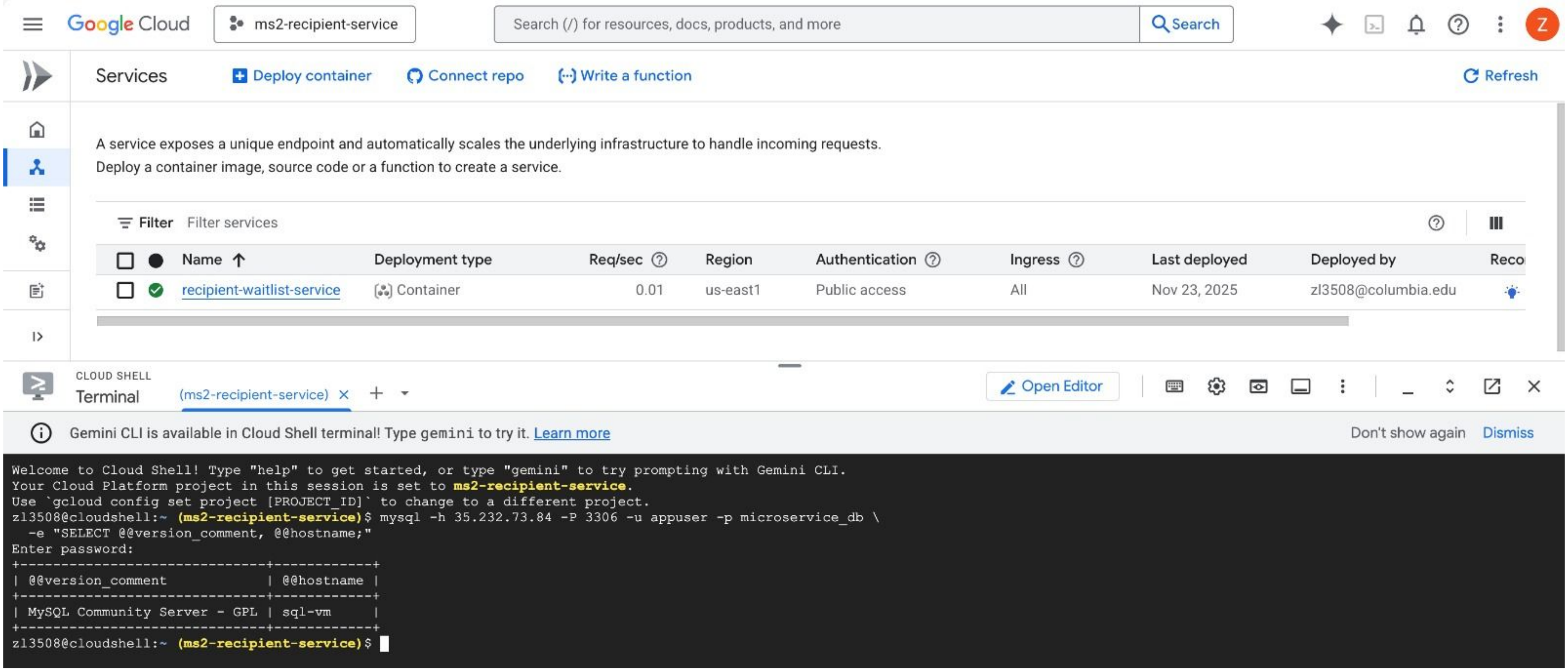
Needs

Column	Type
id	UUID
recipient_id	FK(recipients.id)
organ_type	enum('heart','liver','kidney','lung','pancreas','intestine')
urgency	tinyint unsigned CHECK 1–5
blood_type	enum
status	enum('waiting','matched','removed')
listed_at	datetime
updated_at	datetime

Enums (`models/enums.py`)

- **BloodType**: A+, A-, B+, B-, AB+, AB-, O+, O-
- **OrganType**: heart, liver, kidney, lung, pancreas, intestine
- **CommonStatus**: active, inactive
- **NeedStatus**: waiting, matched, removed

MS2: Cloud Run & sql-vm



MS2: Video Demo

Composite Microservice

The **Composite Microservice** serves as a lightweight routing layer that unifies all API paths from **MS1 (Donor Registry)** and **MS2 (Recipient Waitlist)** under a single service endpoint.

📍 **Composite Service URL (Cloud Run):**

https://composite-service-730071231868.us-central1.run.app/docs

Github Repository:

https://github.com/TheSkyRS/Composite-Service

Composite Microservice: Details

✓ 1. Mirrors all endpoints from MS1 and MS2

All routes from:

- **MS1: Donor Registry**
- **MS2: Recipient Waitlist**

...are duplicated in this composite service.

When a request arrives, the composite service **delegates the call** to the corresponding microservice based on the path.

This provides a **single unified API surface** without modifying any underlying logic.

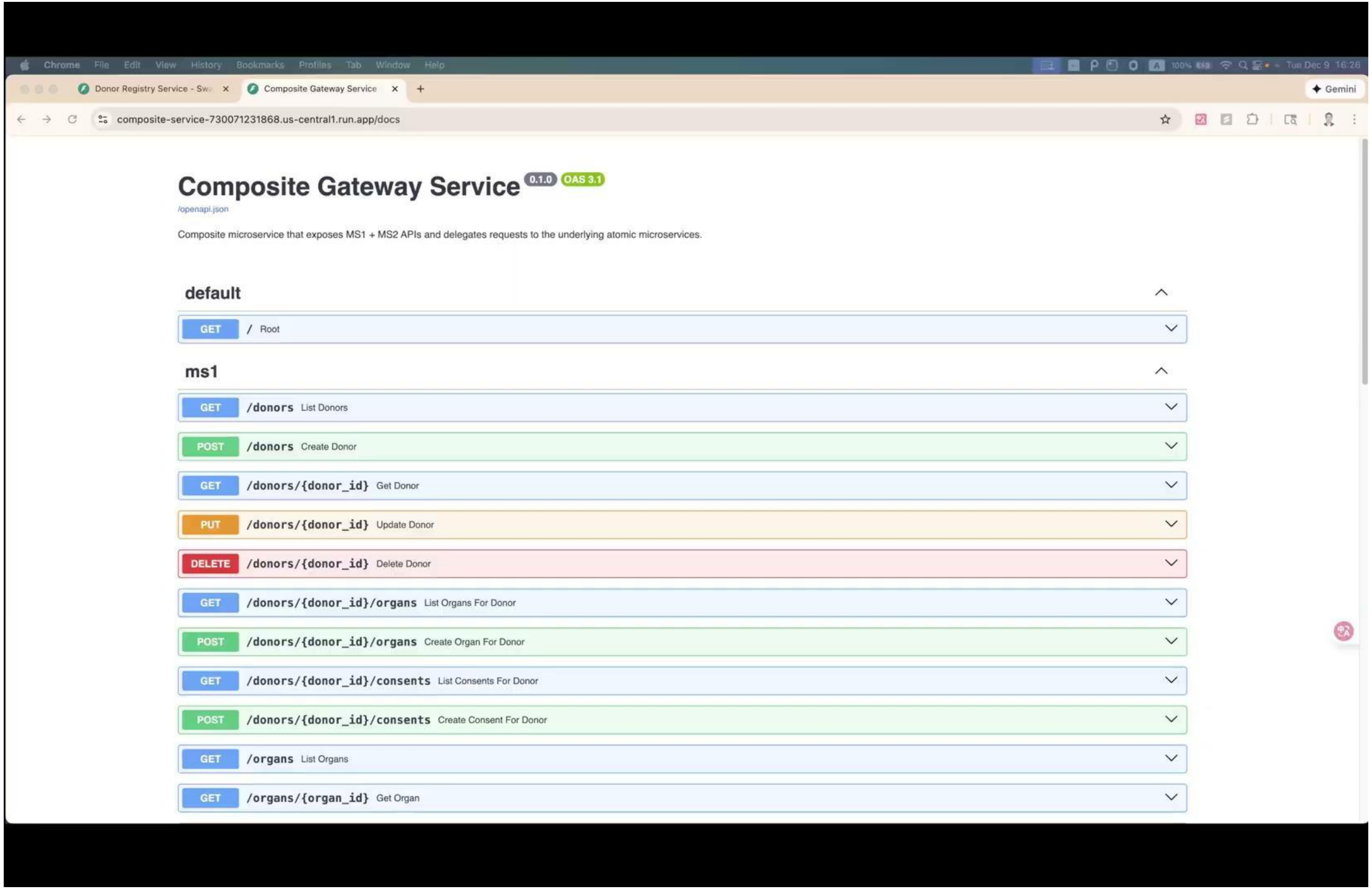
✓ 2. Offers an additional aggregated endpoint

The composite service also includes one extra helper API:

- **Fetch both Needs (from MS2) and Organs (from MS1) in parallel (using threads)**

This endpoint may be useful for **MS3 (Matchmaking)** or for debugging during development.

Composite Microservice: Video Demo 1:32



Microservice 3 - Matchmaking & Notifications Service

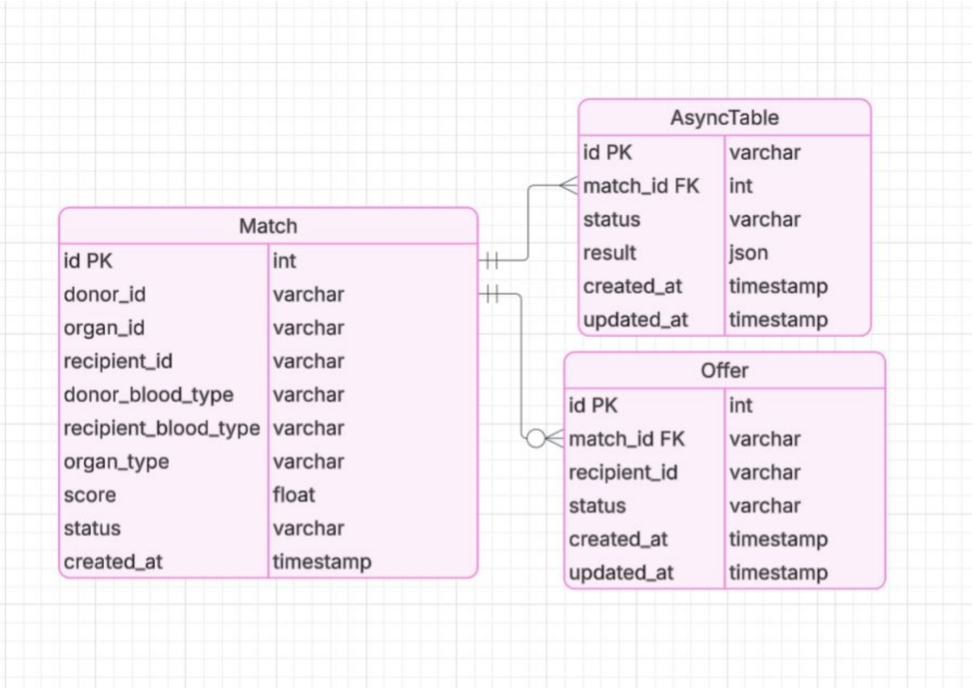
Its main goal is to be able to pair patient’s in need of an organ and with one and triggers a notification cloud function.

Functions:

- Register a patient's health status and needs
- Show if there's an available match
- Create an offer for the patient to accept / deny
- Deletes the organ from the donor and the recipient in need

VM’s URL: http://34.26.41.192:8000/docs

MS3: Data Model: Health, Match, Offer, Async



- **Match**: keeps the hospitable informed on the a donor, recipient match.
- **Offer**: keeps the hospital updated on patient's desire to accept the offer.
- **Async**: Tracks long-running background operations related to a match, including processing status and results.

Cloud Run

- ReDoc: https://matchmaking-service-549668844609.us-central1.run.app/docs

Deployment — Google Cloud Run

gcloud run deploy matchmaking-service

--source .

--region us-central1

--allow-unauthenticated

--set-env-vars DB_HOST=35.243.166.35

--set-env-vars DB_USER=svc_c

--set-env-vars DB_PASSWORD=SvcCpass1!

--set-env-vars DB_NAME=service_c_db

--set-env-vars DB_PORT=3306

```
(venv) amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ gcloud run deploy matchmaking-service \
    --source . \
    --region us-central1 \
    --platform managed
Building using Dockerfile and deploying container to Cloud Run service [matchmaking-service] in project [matchmaking-services] region [us-central1]
✓ Building and deploying... Done.
  ✓ Uploading sources...
  ✓ Building Container... Logs are available at [https://console.cloud.google.com/cloud-build/builds;region=us-central1/b1dc31fa-fd29-4577-80a2-526be36612dd?project=549668844609].
  ✓ Creating Revision...
  ✓ Routing traffic...
Done.
```

MS3: Extra Features Part 1

- ETags enable conditional requests, returning 304 Not Modified when the client's `If-None-Match` value matches the server's ETag, preventing unnecessary data transfer.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ ls
Dockerfile  README.md  docker-compose.yaml  pyproject.toml  requirements.txt  setup.cfg  src  tests  venv
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/offers?limit=10&offset=0"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:50:27 GMT
server: uvicorn
content-length: 1662
content-type: application/json
etag: f5684eee954796420d660babc445aec04abaafbc
link: </offers?limit=10&offset=10>; rel="next"

[{"id":"1","matchId":"38","recipientId":"5dc0cd6a-7350-4cd2-8ce8-9dee43236dd3","status":"pending","createdAt":"2025-11-24T04:34:06","updatedAt":"2025-11-24T04:34:06"},{"id":"2","ma
tchId":"39","recipientId":"ff3b9646-6b5c-4523-9799-d7bf6a5281c9","status":"pending","createdAt":"2025-11-24T04:34:07","updatedAt":"2025-11-24T04:34:07"},{"id":"3","matchId":"40","r
ecipientId":"598f0610-11e7-48d7-ac5f-e4b087d53cc7","status":"pending","createdAt":"2025-11-24T04:34:08","updatedAt":"2025-11-24T04:34:08"},{"id":"4","matchId":"41","recipientId":"8
0cd3fd2-d78c-4635-a102-b737e37ae4be","status":"pending","createdAt":"2025-11-24T04:34:10","updatedAt":"2025-11-24T04:34:10"},{"id":"5","matchId":"42","recipientId":"cfdc0486-5c09-4
3ef-95ba-4f9c874fee24","status":"pending","createdAt":"2025-11-24T04:34:11","updatedAt":"2025-11-24T04:34:11"},{"id":"6","matchId":"43","recipientId":"db4c60cb-30e9-4baa-ae18-9ac01
c13f2c1","status":"pending","createdAt":"2025-11-24T04:34:12","updatedAt":"2025-11-24T04:34:12"},{"id":"7","matchId":"44","recipientId":"c28642e4-af20-45f7-92aa-9c6c1b959593","stat
us":"pending","createdAt":"2025-11-24T04:34:13","updatedAt":"2025-11-24T04:34:13"},{"id":"8","matchId":"45","recipientId":"da3c9172-2fab-4184-9e9b-3a77fc734223","status":"pending",
"createdAt":"2025-11-24T04:34:14","updatedAt":"2025-11-24T04:34:14"},{"id":"9","matchId":"46","recipientId":"0fc078b4-07ea-407a-8c3b-408b65784534","status":"pending","createdAt":"2
025-11-24T04:34:15","updatedAt":"2025-11-24T04:34:15"},{"id":"10","matchId":"47","recipientId":"14f13479-0357-4d40-9ecd-8568cea10468","status":"pending","createdAt":"2025-11-24T04:
34:16","updatedAt":"2025-11-24T04:34:16"}]amy2127@fastapi-openapi-vmcurl -i \aking_and_notifications_service$ curl -i \
  -H "If-None-Match: f5684eee954796420d660babc445aec04abaafbc" \
  "http://localhost:8000/offers?limit=10&offset=0"
HTTP/1.1 304 Not Modified
date: Mon, 08 Dec 2025 06:50:34 GMT
server: uvicorn
```

MS3: Extra Features Part 2

- Query parameters such as `limit` and `offset` allow clients to control pagination manually by specifying how many items to fetch and where in the collection to begin.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/offers?limit=5&offset=10"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:52:27 GMT
server: uvicorn
content-length: 836
content-type: application/json
etag: 2d3b1e3a641acb22950b8a1c2ff84ef94b2fe1d8
link: </offers?limit=5&offset=15>; rel="next"

[{"id":"11","matchId":"48","recipientId":"05037999-340f-44f4-bf17-8abbb79130cc","status":"pending","createdAt":"2025-11-24T04:34:18","updatedAt":"2025-11-24T04:34:18"},{"id":"12","
matchId":"49","recipientId":"82860a69-b8f3-4ad6-96d2-e68c4416d83a","status":"pending","createdAt":"2025-11-24T04:34:19","updatedAt":"2025-11-24T04:34:19"},{"id":"13","matchId":"50"
,"recipientId":"2f747638-88ec-4276-ab23-98b4a0c822cf","status":"pending","createdAt":"2025-11-24T04:34:20","updatedAt":"2025-11-24T04:34:20"},{"id":"14","matchId":"51","recipientId
":"352b1adb-8f76-4323-a5f7-2ad40526aee2","status":"pending","createdAt":"2025-11-24T04:34:21","updatedAt":"2025-11-24T04:34:21"},{"id":"15","matchId":"52","recipientId":"ace43e05-7
4af-4109-af9c-5cd807c98abe","status":"pending","createdAt":"2025-11-24T04:34:22","updatedAt":"2025-11-24T04:34:22"}]amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_servi
ce$ curl -i "http://localhost:8000/matches?limit=3&offset=2"        curl -i "http://localhost:8000/matches?limit=3&offset=2"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:52:32 GMT
server: uvicorn
content-length: 791
content-type: application/json
link: </matches?limit=3&offset=5>; rel="next"

[{"id":"312","donorId":"ab02cd00-90fd-47a8-a201-2efca8051b76","organId":"c4f4809b-b1dd-4339-be89-d48aa264d5b3","recipientId":"00e333af-ffd8-4c49-910d-d7bec50a771d","donorBloodType"
:"O+","recipientBloodType":"O+","organType":"liver","score":1.0,"status":"matched"},{"id":"311","donorId":"bc030bfd-acf5-4cc6-bad9-f92df928211a","organId":"c4cd809c-c9e4-44f7-b031-
e24e452fc6fa","recipientId":"b08ef737-1b48-4586-92e5-7b7407af2939","donorBloodType":"O+","recipientBloodType":"O-","organType":"heart","score":1.0,"status":"matched"},{"id":"310","
donorId":"e169e6c4-01b6-454f-9dfc-444fef9bccd1","organId":"c4c98195-641e-4316-b1de-0b92f3abfd0c","recipientId":"969e7700-0500-4828-b502-82c051e5e149","donorBloodType":"O+","recipie
ntBloodType":"O-","organType":"kidney","score":1.0,"status":"matched"}]amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$
```

MS3: Extra Features Part 3

- Pagination is implemented by returning a `Link` header with the next page's query parameters (e.g., `limit` and `offset`), allowing clients to retrieve results page-by-page.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/offers?limit=5&offset=0"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:53:19 GMT
server: uvicorn
content-length: 831
content-type: application/json
etag: 07d1ad135f1503dfc24fe357d60dd35dfdb6523a
link: </offers?limit=5&offset=5>; rel="next"

[{"id":"1","matchId":"38","recipientId":"5dc0cd6a-7350-4cd2-8ce8-9dee43236dd3","status":"pending","createdAt":"2025-11-24T04:34:06","updatedAt":"2025-11-24T04:34:06"},{"id":"2","ma
tchId":"39","recipientId":"ff3b9646-6b5c-4523-9799-d7bf6a5281c9","status":"pending","createdAt":"2025-11-24T04:34:07","updatedAt":"2025-11-24T04:34:07"},{"id":"3","matchId":"40","r
ecipientId":"598f0610-11e7-48d7-ac5f-e4b087d53cc7","status":"pending","createdAt":"2025-11-24T04:34:08","updatedAt":"2025-11-24T04:34:08"},{"id":"4","matchId":"41","recipientId":"8
0cd3fd2-d78c-4635-a102-b737e37ae4be","status":"pending","createdAt":"2025-11-24T04:34:10","updatedAt":"2025-11-24T04:34:10"},{"id":"5","matchId":"42","recipientId":"cfdc0486-5c09-4
3ef-95ba-4f9c874fee24","status":"pending","createdAt":"2025-11-24T04:34:11","updatedAt":"2025-11-24T04:34:11"}]amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ c
url -i "http://localhost:8000/matches?limit=5&offset=0"              curl -i "http://localhost:8000/matches?limit=5&offset=0"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:53:39 GMT
server: uvicorn
content-length: 1319
content-type: application/json
link: </matches?limit=5&offset=5>; rel="next"

[{"id":"315","donorId":"f0a62189-bf5d-4e11-a331-02a1358c1849","organId":"c8ce7d3e-6883-4508-a4c9-f8ddb6e116e3","recipientId":"5626d569-ad34-4b78-8895-77dfc71737ea","donorBloodType"
:"O+","recipientBloodType":"O+","organType":"kidney","score":1.0,"status":"matched"},{"id":"313","donorId":"589be156-6224-43f3-bce3-aa30b93eb913","organId":"c8a5bb4b-327e-40cc-8c3c
-562a12f2dc37","recipientId":"6afee5ac-560d-4be4-80b6-cd40a950523c","donorBloodType":"B+","recipientBloodType":"B-","organType":"kidney","score":1.0,"status":"matched"},{"id":"312"
,"donorId":"ab02cd00-90fd-47a8-a201-2efca8051b76","organId":"c4f4809b-b1dd-4339-be89-d48aa264d5b3","recipientId":"00e333af-ffd8-4c49-910d-d7bec50a771d","donorBloodType":"O+","recip
ientBloodType":"O+","organType":"liver","score":1.0,"status":"matched"},{"id":"311","donorId":"bc030bfd-acf5-4cc6-bad9-f92df928211a","organId":"c4cd809c-c9e4-44f7-b031-e24e452fc6fa
","recipientId":"b08ef737-1b48-4586-92e5-7b7407af2939","donorBloodType":"O+","recipientBloodType":"O-","organType":"heart","score":1.0,"status":"matched"},{"id":"310","donorId":"e1
69e6c4-01b6-454f-9dfc-444fef9bccd1","organId":"c4c98195-641e-4316-b1de-0b92f3abfd0c","recipientId":"969e7700-0500-4828-b502-82c051e5e149","donorBloodType":"O+","recipientBloodType"
:"O-","organType":"kidney","score":1.0,"status":"matched"}]amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$
```

MS3: Extra Features Part 4

- The service provides Link headers with relative URLs (e.g., </offers?limit=5&offset=5>) that describe relationships like rel="next", enabling clients to navigate between related resources.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/offers?limit=5&offset=0"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:55:13 GMT
server: uvicorn
content-length: 831
content-type: application/json
etag: 07d1ad135f1503dfc24fe357d60dd35dfdb6523a
link: </offers?limit=5&offset=5>; rel="next"

[{"id":"1","matchId":"38","recipientId":"5dc0cd6a-7350-4cd2-8ce8-9dee43236dd3","status":"pending","createdAt":"2025-11-24T04:34:06","updatedAt":"2025-11-24T04:34:06"},{"id":"2","ma
tchId":"39","recipientId":"ff3b9646-6b5c-4523-9799-d7bf6a5281c9","status":"pending","createdAt":"2025-11-24T04:34:07","updatedAt":"2025-11-24T04:34:07"},{"id":"3","matchId":"40","r
ecipientId":"598f0610-11e7-48d7-ac5f-e4b087d53cc7","status":"pending","createdAt":"2025-11-24T04:34:08","updatedAt":"2025-11-24T04:34:08"},{"id":"4","matchId":"41","recipientId":"8
0cd3fd2-d78c-4635-a102-b737e37ae4be","status":"pending","createdAt":"2025-11-24T04:34:10","updatedAt":"2025-11-24T04:34:10"},{"id":"5","matchId":"42","recipientId":"cfdc0486-5c09-4
3ef-95ba-4f9c874fee24","status":"pending","createdAt":"2025-11-24T04:34:11","updatedAt":"2025-11-24T04:34:11"}]amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$
```

MS3: Extra Features Part 5

- Query Parameters Work on Collection Resources
- The API returns a Link header containing **relative paths** (e.g., </offers?limit=5&offset=5>) and a **relationship tag**(rel="next"), demonstrating linked data that guides clients to the next resource page.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/offers?limit=3&offset=2"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 06:56:26 GMT
server: uvicorn
content-length: 499
content-type: application/json
etag: 329aef04f47eda3c7b153a9d6ea90b9a7c0fb6d4
link: </offers?limit=3&offset=5>; rel="next"

[{"id":"3","matchId":"40","recipientId":"598f0610-11e7-48d7-ac5f-e4b087d53cc7","status":"pending","createdAt":"2025-11-24T04:34:08","updatedAt":"2025-11-24T04:34:08"},{"id":"4","ma
tchId":"41","recipientId":"80cd3fd2-d78c-4635-a102-b737e37ae4be","status":"pending","createdAt":"2025-11-24T04:34:10","updatedAt":"2025-11-24T04:34:10"},{"id":"5","matchId":"42","r
ecipientId":"cfdc0486-5c09-43ef-95ba-4f9c874fee24","status":"pending","createdAt":"2025-11-24T04:34:11","updatedAt":"2025-11-24T04:34:11"}]amy2127@fastapi-openapi-vm:~/matchmaking_
and_notifications_service$
```

MS3: Extra Features Part 6

- The POST `/offers` endpoint returns 201 Created along with a `Location` header pointing to the newly created resource, following proper REST conventions.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i -X POST "http://localhost:8000/offers" \
  -H "Content-Type: application/json" \
  -d '{"matchId": "1001", "recipientId": "5dc0cd6a-7350-4cd2-8ce8-9dee43236dd3"}'
HTTP/1.1 201 Created
date: Mon, 08 Dec 2025 07:02:34 GMT
server: uvicorn
content-length: 169
content-type: application/json
location: /offers/292
content-location: /offers/292

{"id":"292","matchId":"1001","recipientId":"5dc0cd6a-7350-4cd2-8ce8-9dee43236dd3","status":"pending","createdAt":"2025-12-08T07:02:35","updatedAt":"2025-12-08T07:02:35"}amy2127@fas
tapi-openapi-vm:~/matchmaking_and_notifications_service$
```

MS3: Extra Features Part 7

- Long-running operations return 202 Accepted with a `Location` link to a task URL, allowing clients to poll the task status until completion.

```
amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i -X POST "http://localhost:8000/matches/1/async"
HTTP/1.1 202 Accepted
date: Mon, 08 Dec 2025 07:05:42 GMT
server: uvicorn
content-length: 69
content-type: application/json
location: /matches/async/tasks/086952ba-e007-43c1-ab76-1b39f6f04ecb
retry-after: 3

{"task_id":"086952ba-e007-43c1-ab76-1b39f6f04ecb","status":"running"}amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i "http://localhost:8000/matches/asyn
c/tasks/918cb2d8-fdbe-4cab-b8ba-c17433ddd3ec"                    curl -i "http://localhost:8000/matches/async/tasks/918cb2d8-fdbe-4cab-b8ba-c17433ddd3ec"
HTTP/1.1 200 OK
date: Mon, 08 Dec 2025 07:05:51 GMT
server: uvicorn
content-length: 234
content-type: application/json

{"id":"918cb2d8-fdbe-4cab-b8ba-c17433ddd3ec","match_id":1,"status":"completed","result":"{\"message\": \"Async processing complete for match 1\", \"match_id\": 1}","created_at":"20
25-12-08T07:04:42","updated_at":"2025-12-08T07:04:48"}amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$
```

Microservice 3 - Matching Algorithm / Future Iterations

- Current
  - The service fetches **all available organs** from MS1 and **all active recipient needs** from MS2.
  - For each organ, it identifies the **first compatible recipient** by comparing organ type and **blood-type compatibility rules**(O → universal donor, AB → universal recipient, etc.).
  - When a compatible match is found, the system **creates a Match record in the SQL database** and assigns a **match score**(currently a simple constant).
  - It also generates a corresponding **Offer** entry so hospitals can later accept or decline the matched organ.
  - After recording the match, the algorithm **consumes the resources** by:
    - deleting the organ from MS1
    - deleting the need from MS2
    - ensuring they cannot be matched again.
  - A **Pub/Sub event is emitted** to notify downstream systems (e.g., notifications, audit log, async processing).
- Future:
  - A production-grade extension of this system would incorporate **Vertex AI** to compute a **machine-learned compatibility score** instead of the current rule-based matching.
  - This could include real-world factors such as tissue compatibility, urgency modeling, geography, donor/recipient prognosis, or learned embeddings from historical match outcomes.
  - The matching engine would call a **Vertex AI Prediction endpoint** to evaluate candidate matches and select the **optimal** instead of the **first compatible** match.

Microservice 3 - Matching Algorithm / Future Iterations

```
(venv) amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i -X POST \
"https://matchmaking-service-549668844609.us-central1.run.app/match/do-match"
HTTP/2 200
content-type: application/json
x-cloud-trace-context: defb7cf215fdf8203309f13c359f01e9;o=1
date: Wed, 10 Dec 2025 10:53:31 GMT
server: Google Frontend
content-length: 6387
```


{"match_count":23,"matches":[{"donor_id":"68e57c5d-ca35-419b-bfba-e7f6de3cc980","organ_id":"425330a1-c9cd-4e5b-9dde-36f4df920860","recipient_id":"29ac3b30-2a10-4dbc-aaf2-d700a0721f
32","donor_blood_type":"A-","recipient_blood_type":"A+","organ_type":"heart","score":1.0,"status":"matched","match_id":355},{"donor_id":"190f1d6c-01db-4c7e-b1fe-41383205850b","orga
n_id":"6c746e31-2b60-4a73-b72c-b7f643751074","recipient_id":"29ac3b30-2a10-4dbc-aaf2-d700a0721f32","donor_blood_type":"A-","recipient_blood_type":"A+","organ_type":"heart","score":
1.0,"status":"matched","match_id":356},{"donor_id":"93ba75cd-df8a-4efc-a1d3-90d6b7bce970","organ_id":"a7edce43-711c-4ca1-93a4-1111a851340f","recipient_id":"4ce32233-f2b7-4712-8e75-
9179853a4dfa","donor_blood_type":"AB+","recipient_blood_type":"AB-","organ_type":"kidney","score":1.0,"status":"matched","match_id":357},{"donor_id":"3297034f-7e76-40c9-af95-60cc56
98ea06","organ_id":"acf887f7-5fc0-465a-a8cc-22bd64d0456f","recipient_id":"72615ca6-60b6-43ff-8c30-4d0720f696cf","donor_blood_type":"AB-","recipient_blood_type":"AB+","organ_type":"
kidney","score":1.0,"status":"matched","match_id":358},{"donor_id":"c8affd15-d800-4d57-9fb3-df8f4d2104be","organ_id":"aed8c033-b0eb-4b3b-b715-b8dc9e7f196d","recipient_id":"cad4bff3
-e229-4490-9a60-b5c7457d1d0a","donor_blood_type":"AB-","recipient_blood_type":"AB-","organ_type":"kidney","score":1.0,"status":"matched","match_id":359},{"donor_id":"3e0e3431-3579-
413e-a012-f0d91ece8bb9","organ_id":"b3a341f7-f712-49c7-bd32-39eb116e7a6b","recipient_id":"8093a0fa-be83-41b3-a993-526b61be6d70","donor_blood_type":"AB-","recipient_blood_type":"AB-
","organ_type":"kidney","score":1.0,"status":"matched","match_id":360},{"donor_id":"eeb752df-2d29-4265-b18a-5a95a69ab80f","organ_id":"b59553db-5ccc-490f-b43c-fae86adffa28","recipie
nt_id":"a3b50384-98ed-4a11-8f7d-f24f9251de55","donor_blood_type":"AB+","recipient_blood_type":"AB-","organ_type":"kidney","score":1.0,"status":"matched","match_id":361},{"donor_id"
:"ad0490e2-acc9-49ee-ac03-289515d48a40","organ_id":"b5c57a19-d432-49d5-bea6-d771fe41a025","recipient_id":"587fcbf6-6712-4835-a97c-3c0f30872f3a","donor_blood_type":"AB+","recipient_
blood_type":"AB+","organ_type":"kidney","score":1.0,"status":"matched","match_id":362},{"donor_id":"8ea3f299-dcf1-442b-943f-81ff9bd07fbd","organ_id":"d4f5e8e8-5ece-4b20-956c-c41874
d33d75","recipient_id":"6a2b5590-59ed-4ea6-92fc-32c4a37a1a01","donor_blood_type":"A+","recipient_blood_type":"A+","organ_type":"kidney","score":1.0,"status":"matched","match_id":36
3},{"donor_id":"2b8282fa-f2f7-47ed-9c80-1c34283f1a09","organ_id":"d6aa6322-82f2-4183-b23a-23da83c046d3","recipient_id":"88369340-e850-437b-9ed1-2ab6178726a2","donor_blood_type":"A-
","recipient_blood_type":"A-","organ_type":"kidney","score":1.0,"status":"matched","match_id":364},{"donor_id":"fbe48d78-40c7-4c99-bef2-0ba5a3862cfb","organ_id":"db4f502f-b19e-4219
-ac0b-bde5c91cfc1a","recipient_id":"e10e33c4-d698-4ca9-a65b-79d617dc946d","donor_blood_type":"A-","recipient_blood_type":"A+","organ_type":"kidney","score":1.0,"status":"matched","
match_id":365},{"donor_id":"11de57a7-d4cb-41f6-a1eb-735537d2c882","organ_id":"dce7b34c-1659-467e-8d67-08110a8a6384","recipient_id":"0b3cc02c-48b2-44af-bc9a-d0848f5d16f4","donor_blo
od_type":"B-","recipient_blood_type":"B+","organ_type":"kidney","score":1.0,"status":"matched","match_id":366},{"donor_id":"dcf4d18c-65a4-4f25-90c4-a3c80dc8b90a","organ_id":"de2e98
55-6133-44d1-8089-db9f33a19572","recipient_id":"39738fd3-4af9-4386-8b40-8c1445a7bd0e","donor_blood_type":"O+","recipient_blood_type":"O-","organ_type":"kidney","score":1.0,"status"
:"matched","match_id":367},{"donor_id":"51d7e816-8127-4338-8fce-7b22f19fcab1","organ_id":"df9c59bd-6cd3-4491-8708-8ebc3feb3238","recipient_id":"9d90bce4-e5f4-430f-a907-a197c34a819f
","donor_blood_type":"A+","recipient_blood_type":"A-","organ_type":"kidney","score":1.0,"status":"matched","match_id":368},{"donor_id":"6b7f120c-4c3a-4e8e-b874-67f8e00b0db4","organ
_id":"e058bbc6-a4f1-470d-b9da-91e31d05dbfc","recipient_id":"7d1835cf-3c21-4870-ad06-cfc7cffcf11f","donor_blood_type":"A-","recipient_blood_type":"A+","organ_type":"kidney","score":
1.0,"status":"matched","match_id":369},{"donor_id":"fdb226d1-6f9f-4d9a-ac2b-47447a84c6af","organ_id":"e05de7b7-27a9-4bd6-b93a-e54fe3c8d6a7","recipient_id":"b43986ea-1309-405c-869e-
281188dfce08","donor_blood_type":"A+","recipient_blood_type":"A+","organ_type":"kidney","score":1.0,"status":"matched","match_id":370},{"donor_id":"893233bc-2ef6-4b09-92b0-a2cced6b
6e09","organ_id":"e1b3265d-bb5a-4e65-86ce-4fee4ff5e3fd","recipient_id":"687c6f83-eb6c-470d-b96c-e2c0558299e3","donor_blood_type":"O+","recipient_blood_type":"O-","organ_type":"kidn
ey","score":1.0,"status":"matched","match_id":371},{"donor_id":"103882e0-8e2c-43bd-a0a2-994befecc09a","organ_id":"e266dace-b612-406c-8095-70a45e90f9f2","recipient_id":"7a6d1796-61c
f-4a53-9528-2b7c524e4d25","donor_blood_type":"B+","recipient_blood_type":"B+","organ_type":"kidney","score":1.0,"status":"matched","match_id":372},{"donor_id":"c4ac511c-e8d7-456b-9
03d-0ee0a614bab4","organ_id":"e3eec008-9fcb-4e42-bff2-85f9cf87ac6b","recipient_id":"3862c710-0363-4605-9f25-50b260537f60","donor_blood_type":"O+","recipient_blood_type":"O-","organ
_type":"liver","score":1.0,"status":"matched","match_id":373},{"donor_id":"9e8a8ea8-3584-4d7c-9e95-fab6e3e465e7","organ_id":"e4d0683d-2265-4782-b11c-14d6ea95c974","recipient_id":"b
23eeffb-6981-4d55-b645-8d833aa03d11","donor_blood_type":"B-","recipient_blood_type":"B-","organ_type":"kidney","score":1.0,"status":"matched","match_id":374},{"donor_id":"fddc631c-
3e5e-4764-bca7-749a637d8040","organ_id":"e52f78d6-3474-4c49-942e-351ddf67bf0e","recipient_id":"5a2c6d7d-8fdc-47c5-87ec-0eb68bc8f143","donor_blood_type":"B+","recipient_blood_type":
"B-","organ_type":"kidney","score":1.0,"status":"matched","match_id":375},{"donor_id":"3d741fae-7084-40dc-bf62-1e53da53f3d9","organ_id":"e56c6dae-63a5-49cb-b034-2d9f13272e0e","reci
pient_id":"0760f897-0d2e-4c5f-87f6-6a2ddc7a3255","donor_blood_type":"B-","recipient_blood_type":"B+","organ_type":"kidney","score":1.0,"status":"matched","match_id":376},{"donor_id
":"0ca604dc-d950-480b-b45b-02cc76b41599","organ_id":"e5cef09c-37c5-4b38-889a-221c6949e6ca","recipient_id":"ef35d181-5805-4c31-9424-828aafacece8","donor_blood_type":"B-","recipient_
blood_type":"B-","organ_type":"kidney","score":1.0,"status":"matched","match_id":377}]}(venv) amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$

Microservice 3 - Matching Algorithm / Future Iterations

- We can actually see the offers being created - triggered by the “do-match” command

```
(venv) amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i \
"https://matchmaking-service-549668844609.us-central1.run.app/offers/354"
HTTP/2 200
content-type: application/json
x-cloud-trace-context: b9622ea2e6ebe00e9e5c259f4a3213cb;o=1
date: Wed, 10 Dec 2025 11:03:36 GMT
server: Google Frontend
content-length: 168

{"id":"354","matchId":"377","recipientId":"ef35d181-5805-4c31-9424-828aafacece8","status":"pending","createdAt":"2025-12-10T10:53:30","updatedAt":"2025-12-10T10:53:30"}(venv) amy21
27@fastapi-openapi-vm:~/matchmaking_and_notifications_service$
```

```
MySQL [service_c_db]> SELECT * FROM offers ORDER BY id DESC LIMIT 20;
+-----+-----------+--------------------------------------+---------+---------------------+---------------------+
| id  | match_id  | recipient_id                         | status  | created_at          | updated_at          |
+-----+-----------+--------------------------------------+---------+---------------------+---------------------+
| 354 | 377       | ef35d181-5805-4c31-9424-828aafacece8 | pending | 2025-12-10 10:53:30 | 2025-12-10 10:53:30 |
| 353 | 376       | 0760f897-0d2e-4c5f-87f6-6a2ddc7a3255 | pending | 2025-12-10 10:53:28 | 2025-12-10 10:53:28 |
| 352 | 375       | 5a2c6d7d-8fdc-47c5-87ec-0eb68bc8f143 | pending | 2025-12-10 10:53:27 | 2025-12-10 10:53:27 |
| 351 | 374       | b23eeffb-6981-4d55-b645-8d833aa03d11 | pending | 2025-12-10 10:53:25 | 2025-12-10 10:53:25 |
| 350 | 373       | 3862c710-0363-4605-9f25-50b260537f60 | pending | 2025-12-10 10:53:24 | 2025-12-10 10:53:24 |
| 349 | 372       | 7a6d1796-61cf-4a53-9528-2b7c524e4d25 | pending | 2025-12-10 10:53:22 | 2025-12-10 10:53:22 |
| 348 | 371       | 687c6f83-eb6c-470d-b96c-e2c0558299e3 | pending | 2025-12-10 10:53:20 | 2025-12-10 10:53:20 |
| 347 | 370       | b43986ea-1309-405c-869e-281188dfce08 | pending | 2025-12-10 10:53:19 | 2025-12-10 10:53:19 |
| 346 | 369       | 7d1835cf-3c21-4870-ad06-cfc7cffcf11f | pending | 2025-12-10 10:53:17 | 2025-12-10 10:53:17 |
| 345 | 368       | 9d90bce4-e5f4-430f-a907-a197c34a819f | pending | 2025-12-10 10:53:16 | 2025-12-10 10:53:16 |
| 344 | 367       | 39738fd3-4af9-4386-8b40-8c1445a7bd0e | pending | 2025-12-10 10:53:14 | 2025-12-10 10:53:14 |
| 343 | 366       | 0b3cc02c-48b2-44af-bc9a-d0848f5d16f4 | pending | 2025-12-10 10:53:12 | 2025-12-10 10:53:12 |
| 342 | 365       | e10e33c4-d698-4ca9-a65b-79d617dc946d | pending | 2025-12-10 10:53:11 | 2025-12-10 10:53:11 |
| 341 | 364       | 88369340-e850-437b-9ed1-2ab6178726a2 | pending | 2025-12-10 10:53:09 | 2025-12-10 10:53:09 |
| 340 | 363       | 6a2b5590-59ed-4ea6-92fc-32c4a37a1a01 | pending | 2025-12-10 10:53:07 | 2025-12-10 10:53:07 |
| 339 | 362       | 587fcbf6-6712-4835-a97c-3c0f30872f3a | pending | 2025-12-10 10:53:06 | 2025-12-10 10:53:06 |
| 338 | 361       | a3b50384-98ed-4a11-8f7d-f24f9251de55 | pending | 2025-12-10 10:53:04 | 2025-12-10 10:53:04 |
| 337 | 360       | 8093a0fa-be83-41b3-a993-526b61be6d70 | pending | 2025-12-10 10:53:02 | 2025-12-10 10:53:02 |
| 336 | 359       | cad4bff3-e229-4490-9a60-b5c7457d1d0a | pending | 2025-12-10 10:53:01 | 2025-12-10 10:53:01 |
| 335 | 358       | 72615ca6-60b6-43ff-8c30-4d0720f696cf | pending | 2025-12-10 10:52:59 | 2025-12-10 10:52:59 |
+-----+-----------+--------------------------------------+---------+---------------------+---------------------+
20 rows in set (0.001 sec)

MySQL [service_c_db]>
```

MS3: Matching + Offers + Cloud Functions Part 1

- **Relationship Between Match and Offer**
  - **A Match represents a successful donor–recipient compatibility** based on organ type, blood type, and medical need.
  - **A Match represents a successful donor–recipient compatibility** based on organ type, blood type, and medical need.’
  - **An Offer is created for the recipient of a Match**, inviting them to accept or decline the organ.
  - **Each Match generates exactly one Offer**, enforced by a **database UNIQUE constraint** on `match_id`.
  - **Matches store medical compatibility details**, while **Offers track the recipient’s response workflow** (pending → accepted/declined).
- **Cloud Function Integration**
  - When a new Match is created, the system **publishes an event to Google Pub/Sub**.
  - A serverless **Cloud Run “handle-event” function** automatically triggers on this Pub/Sub message.
  - The Cloud Function can **notify downstream systems**, log the match, send alerts, or trigger additional workflows.
  - The Cloud Function can **notify downstream systems**, log the match, send alerts, or trigger additional workflows.

MS3: Matching + Offers + Cloud Functions Part 2

- On the Server Side - Pub triggered:

```
DEBUG DB SETTINGS: {'DB_HOST': '35.243.166.35', 'DB_USER': 'svc_c', 'DB_NAME': 'service_c_db', 'DB_PORT': '3306'}
DEBUG DB SETTINGS: {'DB_HOST': '35.243.166.35', 'DB_USER': 'svc_c', 'DB_NAME': 'service_c_db', 'DB_PORT': '3306'}
Published event ID: 17442667558877791
MATCH EVENT RECEIVED: {'match_id': 332, 'donor_id': 'dc5eca22-996f-4d31-876f-1ce652eab3c4', 'organ_id': 'd0400e1d-6ce4-49ed-8638-e94f445d8a8d', 'recipient_id': '4a7033be-013c-4df4-
97e3-95bf721bb918', 'organ_type': 'kidney', 'message': 'New donor-recipient match created'}
DEBUG DB SETTINGS: {'DB_HOST': '35.243.166.35', 'DB_USER': 'svc_c', 'DB_NAME': 'service_c_db', 'DB_PORT': '3306'}
DEBUG DB SETTINGS: {'DB_HOST': '35.243.166.35', 'DB_USER': 'svc_c', 'DB_NAME': 'service_c_db', 'DB_PORT': '3306'}
Published event ID: 17442348216025153
MATCH EVENT RECEIVED: {'match_id': 333, 'donor_id': '15e13663-df93-47eb-84be-342981e8c163', 'organ_id': 'd096209d-b6e8-4e20-ae53-db78ecdbb1ef', 'recipient_id': '3e6c43e7-420f-41c7-
95ed-fde1caeeb966', 'organ_type': 'kidney', 'message': 'New donor-recipient match created'}
INFO:     127.0.0.1:43086 - "POST /match/do-match HTTP/1.1" 200 OK
```

- On the terminal side:

```
(venv) amy2127@fastapi-openapi-vm:~/matchmaking_and_notifications_service$ curl -i -X POST "http://localhost:8000/match/do-match"
HTTP/1.1 200 OK
date: Tue, 09 Dec 2025 10:55:06 GMT
server: uvicorn
content-length: 5002
content-type: application/json
```

{"match_count":18,"matches":[{"donor_id":"9d847a56-c0f8-4ee4-a654-f8e72377dfe5","organ_id":"63d71534-761e-4823-bc5e-333d82e903ef","recipient_id":"d339d7ee-d2cc-44f4-8c6a-a67a9415ea
eb","donor_blood_type":"AB+","recipient_blood_type":"AB+","organ_type":"liver","score":1.0,"status":"matched","match_id":316},{"donor_id":"92af5573-dad0-43f8-bcd8-cedd50597ecf","or
gan_id":"7130d997-25fd-409a-9d10-3101f1f99f4d","recipient_id":"7b89b920-23fd-49b0-805c-1e4fac36bb7e","donor_blood_type":"AB+","recipient_blood_type":"AB+","organ_type":"heart","sco
re":1.0,"status":"matched","match_id":317},{"donor_id":"b1dc42f4-5f69-4658-a021-90125e5ecf40","organ_id":"98d84647-6984-4bef-b5ac-b38f6eefdbac","recipient_id":"e9260356-30c3-4175-a
3e9-531bbc5dbd40","donor_blood_type":"AB+","recipient_blood_type":"AB-","organ_type":"kidney","score":1.0,"status":"matched","match_id":318},{"donor_id":"9c917d5b-c494-42f6-abab-8e
cf0fd5bc28","organ_id":"a3b62a67-ca3f-4137-b719-59eb13850290","recipient_id":"4724eebf-c0c0-402c-a621-1a1c113cc258","donor_blood_type":"AB+","recipient_blood_type":"AB-","organ_typ

MS3: Note on Swagger

API-First → Code-First Workflow Notes

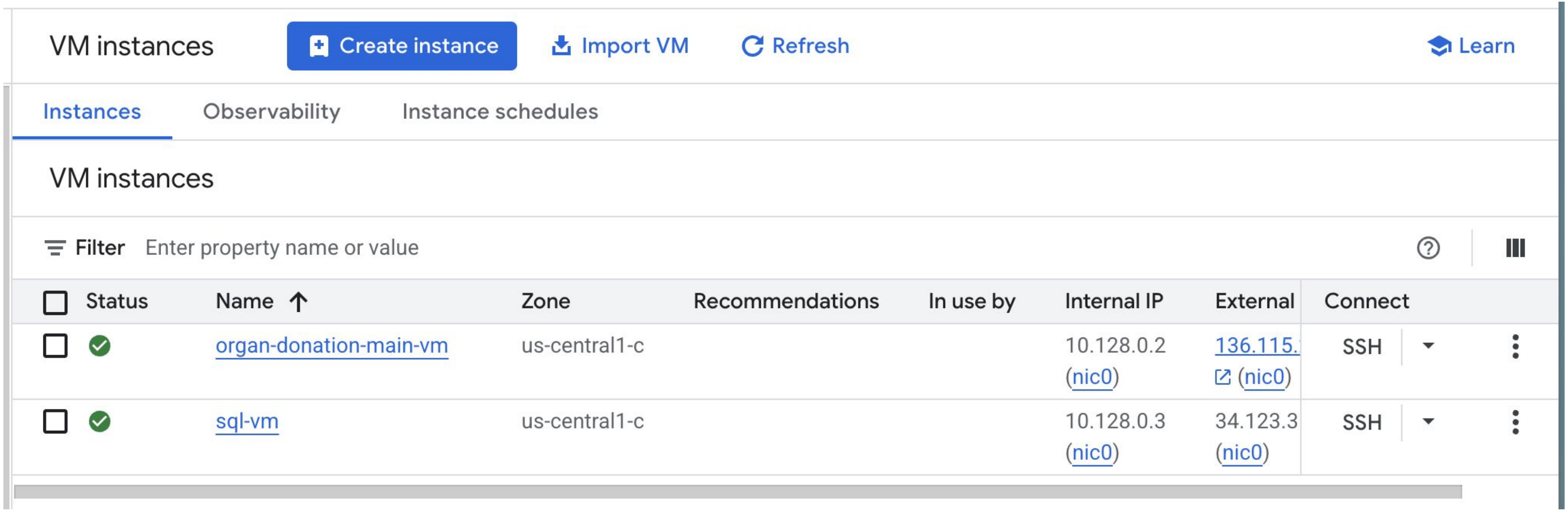
This project originally began using a **Swagger / OpenAPI-first** approach. Indeed the first couple Sprints operated in this way. However, as the project grew and some more sophistcated functionality was needed, it became clear that implementing the service directly in code was a more flexible way to expand functionality, experiment with logic, and iterate quickly. Because of this, I transitioned partially to a code-first workflow, building out FastAPI routes directly and allowing the implementation to drive the final API shape.

After completing the core functionality (Matches, Offers, Composite services, async tasks, etc.), I went back and **updated the OpenAPI YAML file to match the final implementation**. The YAML now serves primarily as:

- A reference document
- A complete description of all active endpoints
- A way to expose Swagger UI at `/docs`

The original API-first drafts and earlier versions of the spec are preserved in the `archive/` directory for historical reference.

VM’s Screenshot



Databases

db > schema.sql

```sql
-- Donor Registry - Relational Schema

CREATE DATABASE IF NOT EXISTS donor_registry
  CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci;
USE donor_registry;

-- Reference/lookup: Hospitals
CREATE TABLE IF NOT EXISTS hospitals (
  hospital_id     BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  name            VARCHAR(200) NOT NULL,
  city            VARCHAR(120) NULL,
  state           VARCHAR(80)  NULL,
  created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
) ENGINE=InnoDB;

-- Reference/lookup: Organ types
CREATE TABLE IF NOT EXISTS organ_types (
  organ_type_id   BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  name            VARCHAR(80) NOT NULL,
  description     TEXT NULL,
  created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  UNIQUE KEY uk_organ_types_name (name)
) ENGINE=InnoDB;

-- Donors
CREATE TABLE IF NOT EXISTS donors (
  donor_id        BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  first_name      VARCHAR(120) NOT NULL,
  last_name       VARCHAR(120)  NOT NULL,
  dob             DATE          NULL,
  gender          ENUM('M','F','O') NULL,
  blood_type      ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-') NOT NULL,
  hla_type        VARCHAR(200)  NULL,
  comorbidities   TEXT          NULL,
  city            VARCHAR(120)  NULL,
  state           VARCHAR(80)   NULL,
  hospital_id     BIGINT UNSIGNED NULL,
  created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  CONSTRAINT fk_donors_hospital
    FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
    ON UPDATE CASCADE ON DELETE SET NULL,
```

```sql
USE donor_registry;

-- Organ types
INSERT INTO organ_types (name, description) VALUES
  ('Kidney',''),
  ('Liver',''),
  ('Heart','');

-- Hospitals
INSERT INTO hospitals (name, city, state) VALUES
  ('Columbia University Irving Medical Center','New York','NY'),
  ('NYU Langone Health','New York','NY');

-- Donors
INSERT INTO donors (first_name,last_name,dob,gender,blood_type,hla_type,city,state,hospital_id)
VALUES
  ('Ravi','Sharma','1988-05-12','M','O+','A*01:01 B*08:01','New York','NY',1),
  ('Neha','Kapoor','1990-11-23','F','A-','A*02:01 B*07:02','New York','NY',2);

-- Recipients
INSERT INTO recipients (first_name,last_name,dob,gender,blood_type,hla_type,needed_organ_id,urgency,waitlist_date,city,state,hospital_id)
VALUES
  ('Arjun','Patel','1995-02-08','M','O+','A*01:01 B*08:01', (SELECT organ_type_id FROM organ_types WHERE name='Kidney'), 9, CURRENT_DATE, 'New Y
  ('Meera','Iyer','1986-07-15','F','A-','A*02:01 B*07:02', (SELECT organ_type_id FROM organ_types WHERE name='Liver'), 6, CURRENT_DATE, 'New Yo

-- Donations (link donor -> organ)
INSERT INTO donations (donor_id, organ_type_id, donation_date, status, health_rating, hospital_id)
VALUES
  (1, (SELECT organ_type_id FROM organ_types WHERE name='Kidney'), CURRENT_DATE, 'Available', 8, 1),
  (2, (SELECT organ_type_id FROM organ_types WHERE name='Liver'), CURRENT_DATE, 'Available', 7, 2);

-- Matches (example proposed matches with a score)
INSERT INTO matches (donation_id, recipient_id, match_score, match_status)
VALUES
  (1, 1, 92.500, 'Proposed'),
  (2, 2, 76.300, 'Proposed');
  ⌘L to chat, ⌘K to generate
```

Authentication using Firebase

✅ Support for OAuth2/OIDC login using Google

✅ After login, generate own JWT token (include custom field: “roles”)

User / Roles


```json
{
  "email": "yl5763@columbia.edu",
  "uid": "4BbAnDoKREZ3K144q6vpOxwNkwM2",
  "roles": [
    "ms1admin",
    "ms2admin",
    "ms3admin"
  ]
}
```

Authentication using Firebase

JWT

JWT Decoder JWT Encoder

Paste a JWT below that you'd like to decode, validate, and verify.

Generate example

ENCODED VALUE

Enable auto-focus

JSON WEB TOKEN (JWT)

COPY CLEAR

Valid JWT

Fix public key input errors to verify signature.

eyJhbGciOiJSUzI1NiIsImtpZCI6Ijk1MTg5MTkxMTA3NjA1NDM0NGUxNWUyNTY0MjViYjQyNWVlYjNhNWMiLCJ0eXAiOiJKV1QifQ.eyJuYW1lIjoiWW9uZ2hhbyBMaW4iLCJwaWN0dXJlIjoiaHR0cHM6Ly9saDMuZ29vZ2xldXNlcmNvbnRlbnQuY29tL2EvQUNnOG9jTG5HcVdERDhxbl9sSlR6SnVwVWstOWZ2eW1nWVpKam9VZllJWEs3MlhwQjFrejRxdz1zOTYtYyIsInJvbGVzIjpbIm1zMWFkbWluIiwibXMyYWRtaW4iLCJtczNhZG1pbiJdLCJpc3MiOiJodHRwczovL3NlY3VyZXRva2VuLmdvb2dsZS5jb20vY2xvdWQtY29tcHV0aW5nLTQ3ODQyMCIsImF1ZCI6ImNsb3VkLWNvbXB1dGluZy00Nzg0MjAiLCJhdXRoX3RpbWUiOjE3NjUxNTMwMjEsInVzZXJfaWQiOiI0QmJBbkRvS1JFWjNLMTQ0cTZ2cE94d05rd00yIiwic3ViIjoiNEJiQW5Eb0tSRVozSzE0NHE2dnBPeHdOa3dNMiIsImlhdCI6MTc2NTMwMDA5NiwiZXhwIjoxNzY1MzAzNjk2LCJlbWFpbCI6InlsNTc2M0Bjb2x1bWJpYS5lZHUiLCJlbWFpbF92ZXJpZmllZCI6dHJ1ZSwiZmlyZWJhc2UiOnsiaWRlbnRpdGllcyI6eyJnb29nbGUuY29tIjpbIjEwMzU0ODY2OTc5MDk1MTE2OTMxNCJdLCJlbWFpbCI6WyJ5bDU3NjNAY29sdW1iaWEuZWR1Il19LCJzaWduX2luX3Byb3ZpZGVyIjoiZ29vZ2xlLmNvbSJ9fQ.BXCYIWoGe15O4AmIcy2P4xualvcRuSeZ8Gwm6j3_tRwAe6n33bwY1uQu3zkimyggnOO1ww9jkNvmYoOCZmdURphqSiKELEM7khp0Jb8V-2DslsID-vnP5Kx7tuh8Lia_oK26y_SRHkJ1VIcFaw8aEzV-_RkY-ceX50crIMvxnWfrttr50y_q139ES7aKkAnJRmh7k86-VJJbJ8-oN5wMPUirS_wCcspKnCUMrl9jXboM13fF4DcQjsGAPu9CH3oVlyGJh4gNuNT6ryYzILtZzhcUQM4KvyQQN-HomhI_wlrnfa2DefqwD1kvh1brGTurwC7b9benBsMxDL7rnC0SGg

:authority	cloud-gateway-9be1k7ss.uc.gateway.dev
:method	GET
:path	/ms1/donors/05538195-b559-4814-bbeb-9d4303d2b085/organs
:scheme	https
Accept	*/*
Accept-Encoding	gzip, deflate, br, zstd
Accept-Language	zh-CN,zh;q=0.9,en;q=0.8,zh-TW;q=0.7
Authorization	Bearer eyJhbGciOiJSUzI1NiIsImtpZCI6Ijk1MTg5MTkxMTA3NjA1NDM0NGUxNWUyNTY0MjViYjQyNWVlYjNhNWMiLCJ0eXAiOiJKV1QifQ.eyJuYW1lIjoiWW9uZ2hhbyBMaW4iLCJwaWN0dXJlIjoiaHR0cHM6Ly9saDMuZ29vZ2xldXNlcmNvbnRlbnQuY29tL2EvQUNnOG9jTG5HcVdERDhxbl9sSlR6SnVwVWstOWZ2eW1nWVpKam9VZllJWEs3MlhwQjFrejRxdz1zOTYtYyIsInJvbGVzIjpbIm1zMWFkbWluIiwibXMyYWRtaW4iLCJtczNhZG1pbiJdLCJpc3MiOiJodHRwczovL3NlY3VyZXRva2VuLmdvb2dsZS5jb20vY2xvdWQtY29tcHV0aW5nLTQ3ODQyMCIsImF1ZCI6ImNsb3VkLWNvbXB1dGluZy00Nzg0MjAiLCJhdXRoX3RpbWUiOjE3NjUxNTMwMjEsInVzZXJfaWQiOiI0QmJBbkRvS1JFWjNLMTQ0cTZ2cE94d05rd00yIiwic3ViIjoiNEJiQW5Eb0tSRVozSzE0NHE2dnBPeHdOa3dNMiIsImlhdCI6MTc2NTMwMDA5NiwiZXhwIjoxNzY1MzAzNjk2LCJlbWFpbCI6InlsNTc2M0Bjb2x1bWJpYS5lZHUiLCJlbWFpbF92ZXJpZmllZCI6dHJ1ZSwiZmlyZWJhc2UiOnsiaWRlbnRpdGllcyI6eyJnb29nbGUuY29tIjpbIjEwMzU0ODY2OTc5MDk1MTE2OTMxNCJdLCJlbWFpbCI6WyJ5bDU3NjNAY29sdW1iaWEuZWR1Il19LCJzaWduX2luX3Byb3ZpZGVyIjoiZ29vZ2xlLmNvbSJ9fQ.BXCYIWoGe15O4AmIcy2P4xualvcRuSeZ8Gwm6j3_tRwAe6n33bwY1uQu3zkimyggnOO1ww9jkNvmYoOCZmdURphqSiKELEM7khp0Jb8V-2DslsID-vnP5Kx7tuh8Lia_oK26y_SRHkJ1VIcFaw8aEzV-_RkY-ceX50crIMvxnWfrttr50y_q139ES7aKkAnJRmh7k86-VJJbJ8-oN5wMPUirS_wCcspKnCUMrl9jXboM13fF4DcQjsGAPu9CH3oVlyGJh4gNuNT6ryYzILtZzhcUQM4KvyQQN-HomhI_wlrnfa2DefqwD1kvh1brGTurwC7b9benBsMxDL7rnC0SGg
Origin	https://storage.googleapis.com

DECODED HEADER

JSON CLAIMS TABLE COPY


```json
{
  "alg": "RS256",
  "kid": "951891911076054344e15e256425bb425eeb3a5c",
  "typ": "JWT"
}
```


DECODED PAYLOAD

JSON CLAIMS TABLE COPY


```json
{
  "name": "Yonghao Lin",
  "picture": "https://lh3.googleusercontent.com/a/ACg8ocLnGqWDD8qn_lJTzJupUk-9fvymgYZJjoUfYIXK72XpB1kz4qw=s96-c",
  "roles": [
    "ms1admin",
    "ms2admin",
    "ms3admin"
  ],
  "iss": "https://securetoken.google.com/cloud-computing-478420",
  "aud": "cloud-computing-478420",
  "auth_time": 1765153021,
  "user_id": "4BbAnDoKREZ3K144q6vpOxwNkwM2",
  "sub": "4BbAnDoKREZ3K144q6vpOxwNkwM2",
  "iat": 1765300096,
  "exp": 1765303696,
  "email": "yl5763@columbia.edu",
  "email_verified": true,
  "firebase": {
    "identities": {
      "google.com": [
        "103548669790951169314"
      ],
      "email": [
        "yl5763@columbia.edu"
      ]
    },
    "sign_in_provider": "google.com"
  }
}
```

Google API gateway to verify the JWT token

1. When missing JWT token

```
✘ theskyrs@Shirito-MacBook ~ curl https://cloud-gateway-9be1k7ss.uc.gateway
.dev/ms1/donors/05538195-b559-4814-bbeb-9d4303d2b085
{"code":401,"message":"Jwt is missing"}
```

2. When JWT token is modified

```
theskyrs@Shirito-MacBook ~ curl https://cloud-gateway-9be1k7ss.uc.gateway.dev/ms1/donors/05538195-b559-4814-bbeb-9d4303d2b085 -H "Authoriza
tion: Bearer eyJhbGciOiJSUzI1NiIsImtpZCI6Ijk1MTg5MTkxMTA3NjA1NDM0NGUxNWUyNTY0MjViYjQyNWVlYjNhNWMiLCJ0eXAiOiJKV1QifQ.eyJuYW1lIjoiWW9uZ2hhbyBMaW4i
LCJwaWN0dXJlIjoiaHR0cHM6Ly9saDMuZ29vZ2xldXNlcmNvbnRlbnQuY29tL2EvQUNnOG9jTG5HcVdERDhxbl9sSlR6SnVwVWstOWZ2eW1nWVpKam9VZllJWEs3MlhwQjFrejRxdz1zOTYt
YyIsInJvbGVzIjpbIm1zMWFkbWluIiwibXMyYWRtaW4iLCJtczNhZG1pbiJdLCJpc3MiOiJodHRwczovL3NlY3VyZXRva2VuLmdvb2dsZS5jb20vY2xvdWQtY29tcHV0aW5nLTQ3ODQyMCIs
ImF1ZCI6ImNsb3VkLWNvbXB1dGluZy00Nzg0MjAiLCJhdXRoX3RpbWUiOjE3NjUxNTMwMjEsInVzZXJfaWQiOiI0QmJBbkRvS1JFWjNLMTQ0cTZ2cE94d05rd00yIiwic3ViIjoiNEJiQW5E
b0tSRVozSzE0NHE2dnBPeHd0a3dNMiIsImlhdCI6MTc2NTMwMDA5NiwiZXhwIjoxNzY1MzAzNjk2LCJlbWFpbCI6InlsNTc2M0Bjb2x1bWJpYS5lZHUiLCJlbWFpbF92ZXJpZmllZCI6dHJ1
ZSwiZmlyZWJhc2UiOnsiaWRlbnRpdGllcyI6eyJnb29nbGUuY29tIjpbIjEwMzU0ODY2OTc5MDk1MTE2OTMxNCJdLCJlbWFpbCI6WyJ5bDU3NjNAY29sdW1iaWEuZWR1Il19LCJzaWduX2lu
X3Byb3ZpZGVyIjoiZ29vZ2xlLmNvbSJ9fQ.BXCYIWoGe15O4AmIcy2P4xualvcRuSeZ8Gwm6j3_tRwAe6n33bwY1uQu3zkimyggnOO1ww9jkNvmYoOCZmdURphqSiKELEM7khp0Jb8V-2Dsl
sID-vnP5Kx7tuh8Lia_oK26y_SRHkJ1VIcFaw8aEzV-_RkY-ceX50crIMvxnWfrttr50y_q139ES7aKkAnJRmh7k86-VJJbJ8-oN5wMPUirS_wCcspKnCUMrl9jXboM13fF4DcQjsGAPu9CH
3oVlyGJh4gNuNT6ryYzILtZzhcUQM4KvyQQN-HomhI_wlrnfa2DefqwD1kvh1brGTurwC7b9benBsMxDL7rnC0SGas"
{"message":"Jwt verification fails","code":401}
```

Authorization using RBAC Service

When JWT token is verified by API-Gateway, we need to check the “roles” of user.

We built a new microservice called RBAC service for Authorization, it will based on the user’s roles to determine whether we should delegate the request to the underlying MS1/MS2/MS3.

Also, the RBAC service will do CORS response and expose ETag/Location headers.

Authorization using RBAC Service

1. Have MS1admin role and request to access MS1 (this is allowed)

```
theskyrs@Shirito-MacBook ~ curl https://cloud-gateway-9be1k7ss.uc.gateway.dev/ms1/donors/05538195-b559-4814-bbeb-9d4303d2b085 -H "
Authorization: Bearer eyJhbGciOiJSUzI1NiIsImtpZCI6Ijk1MTg5MTkxMTA3NjA1NDM0NGUxNWUyNTY0MjViYjQyNWVlYjNhNWMiLCJ0eXAiOiJKV1QifQ.eyJuYW1lIj
oiWW9uZ2hhbyBMaW4iLCJwaWN0dXJlIjoiaHR0cHM6Ly9saDMuZ29vZ2xldXNlcmNvbnRlbnQuY29tL2EvQUNnOG9jTG5HcVdERDhxbl9sSlR6SnVwVWstOWZ2eW1nWVpKam9VZ
llJWEs3MlhwQjFrejRxdz1zOTYtYyIsInJvbGVzIjpbIm1zMWFkbWluIl0sImlzcyI6Imh0dHBzOi8vc2VjdXJldG9rZW4uZ29vZ2xlLmNvbS9jbG91ZC1jb21wdXRpbmctNDc4
NDIwIiwiYXVkIjoiY2xvdWQtY29tcHV0aW5nLTQ3ODQyMCIsImF1dGhfdGltZSI6MTc2NTMwMTgxOSwidXNlcl9pZCI6IjRCYkFuRG9LUkVaM0sxNDRxNnZwT3h3Tmt3TTIiLCJ
zdWIiOiI0QmJBbkRvS1JFWjNLMTQ0cTZ2cE94d05rd00yIiwiaWF0IjoxNzY1MzAxODIwLCJleHAiOjE3NjUzMDU0MjAsImVtYWlsIjoieWw1NzYzQGNvbHVtYmlhLmVkdSIsIm
VtYWlsX3ZlcmlmaWVkIjp0cnVlLCJmaXJlYmFzZSI6eyJpZGVudGl0aWVzIjp7Imdvb2dsZS5jb20iOlsiMTAzNTQ4NjY5NzkwOTUxMTY5MzE0Il0sImVtYWlsIjpbInlsNTc2M
0Bjb2x1bWJpYS5lZHUiXX0sInNpZ25faW5fcHJvdmlkZXIiOiJnb29nbGUuY29tIn19.RdWNwbFhd1FD8IRQrQcio1N-EyxkJDa9mt2VK2tXNOLLXbCjSME-UzALGzCT3oOw7vY
VZLWiojykdY1S1W_aXobqt7Ny8_bHbaxs990-n3_E_KfcsgnwuNNM7HAeLbVjaomeOdqWm8X4V5AdDfLcJLzN6jWnLJ6b7xHF3Nxi_2bMt4VTJQxs4w6G75xCyZ9IBeO0vEnhW3
QisNFYo8VDl_dM0RRv9S8bDehlolBvU1JfZlGTCZ5flz3TcRc8IJJL5zZNMpyR914-r_AUJRQv6QNswsqxZ_5MgUwgVOhMI3DCsYxSIPUuqnwofAyPx2XihMBICCL9Xm4ChoGV0
VHerQ"
{"full_name":"Donor 13","dob":"2021-04-07","blood_type":"A-","status":"active","id":"05538195-b559-4814-bbeb-9d4303d2b085","created_at"
:"2025-11-24T03:06:23","updated_at":"2025-11-24T03:06:23"}%
```

2. Only Have MS1admin role but request to access MS2 (this is not allowed)

```
theskyrs@Shirito-MacBook ~ curl https://cloud-gateway-9be1k7ss.uc.gateway.dev/ms2/needs -H "Authorization: Bearer eyJhbGciOiJSUzI1
NiIsImtpZCI6Ijk1MTg5MTkxMTA3NjA1NDM0NGUxNWUyNTY0MjViYjQyNWVlYjNhNWMiLCJ0eXAiOiJKV1QifQ.eyJuYW1lIjoiWW9uZ2hhbyBMaW4iLCJwaWN0dXJlIjoiaHR0
cHM6Ly9saDMuZ29vZ2xldXNlcmNvbnRlbnQuY29tL2EvQUNnOG9jTG5HcVdERDhxbl9sSlR6SnVwVWstOWZ2eW1nWVpKam9VZllJWEs3MlhwQjFrejRxdz1zOTYtYyIsInJvbGV
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{"detail":"ms2admin role required to access this resource"}%
```
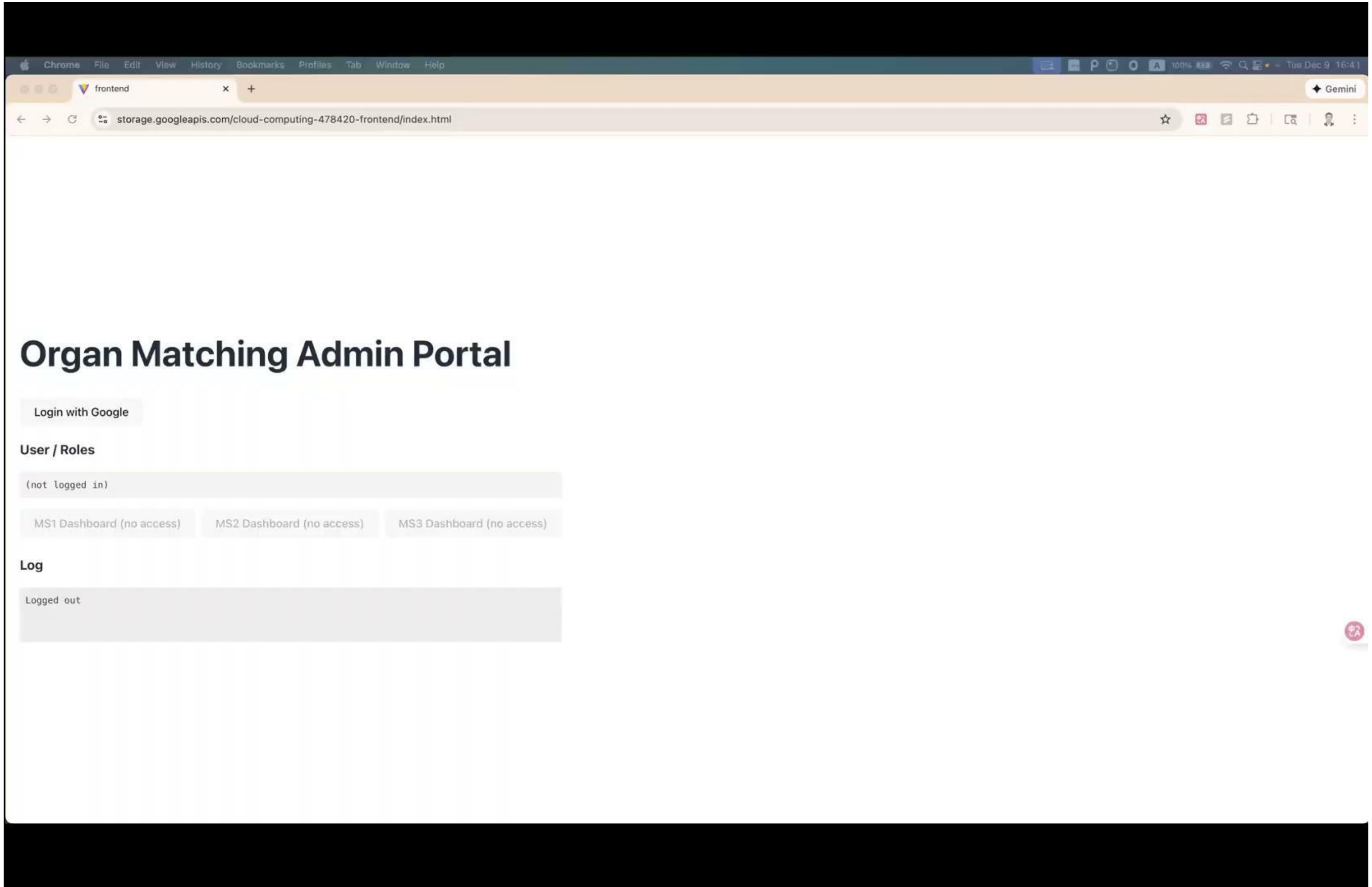

Authentication and Authorization: Video Demo 3:26



Browser/web application

React built frontend, deployed on Cloud Storage.

Access URL: https://storage.googleapis.com/cloud-computing-478420-frontend/index.html

Github Repository: https://github.com/TheSkyRS/4153-SimpleFrontEnd

Conclusion

System Requirements

- 6 GitHub repositories (3 atomic MS, composite MS, DB model, web UI). ✅
- Full CRUD for all resources across all microservices. ✅
- OpenAPI specs for all services. ✅

Cloud Deployment & Databases — Completed

- At least one MS on Cloud Compute. ✅
- At least one MS on Cloud Run. ✅
- MySQL VM used by MS2. ✅
- Cloud SQL used by MS1 and MS3. ✅
- Browser UI deployed on Cloud Storage. ✅

Conclusion

Advanced Features — Completed

- eTag processing (MS1 & MS3). ✅
- Query parameters on all collections (MS1 & MS3). ✅
- Pagination (MS1 & MS3). ✅
- Linked data + relative paths (MS1 & MS3). ✅
- 201 Created + Location header (MS1 & MS3). ✅
- 202 Accepted + async + polling (MS3). ✅
- Composite MS with parallel threads. ✅
- Logical foreign key constraints. (MS3) ✅
- Simple web UI which deployed via Cloud Storage. ✅
- OAuth2/OIDC login + custom JWT + RBAC. ✅
- Cloud Function triggered by Pub/Sub (MS3). ✅

PHOTO: DARCY PETERKA

Thank you!